

Pattern	Question	Link 1	Link 2	Link 3
1. Pattern: Two Pointers				
Two with Target Sum (easy)				
Remove Duplicates (easy)	https://leetcode.com/problems/remove-duplicates-from-sorted-list/		https://leetcode.com/problems/remove-duplicates-from-sorted-array-ii/	
Squaring a Sorted Array (easy)	https://leetcode.com/problems/squares-of-a-sorted-array/			
Triplet Sum to Zero (medium)	https://leetcode.com/problems/3sum/			
Triplet Sum Close to Target (medium)	https://leetcode.com/problems/3sum-closest/			
Triplets with Smaller (medium)	https://www.geeksforgeeks.org/problems/least-triplets-with-sum-smaller-than-x554971			
Subarray with Product Less than a Target (medium)	https://leetcode.com/problems/subarray-product-less-than-k/			
Dutch National Flag Problem (medium)	https://leetcode.com/problems/part-colors/description/			
Problem Challenge 1: Quadruple Sum to Target (medium)	https://leetcode.com/problems/4sum/			
Problem Challenge 2: Comparing Strings containing Backspaces (medium)	https://leetcode.com/problems/backspace-string-compare/			
Problem Challenge 3: Minimum Window Sort (medium)	https://leetcode.com/problems/shortest-unsorted-continuous-subarray/		https://www.ideserve.co.in/learn/minimum-length-subarray-sorting-which-results-in-sorted-array	
2. Pattern: Fast & Slow pointers				
Linked List Cycle (easy)	https://leetcode.com/problems/linked-list-cycle/			
Start of Linked List Cycle (medium)	https://leetcode.com/problems/linked-list-cycle-ii/			
Happy Number (medium)	https://leetcode.com/problems/happy-number/			
Find the Duplicate Number (hard)	https://leetcode.com/problems/find-the-duplicate-number/description/			
Start of the Linked List (easy)	https://leetcode.com/problems/middle-of-the-linked-list/			
Problem Challenge 1: Palindrome Linked List (medium)	https://leetcode.com/problems/palindrome-linked-list/			
Problem Challenge 2: Rearrange a Linked List (medium)	https://leetcode.com/problems/reorder-list/			
Problem Challenge 3: Cycle in a Circular Array (hard)	https://leetcode.com/problems/circular-array-loop/			
3. Pattern: Sliding Window				
Maximum Sum Subarray of Size K (easy)	https://www.geeksforgeeks.org/problems/max-sum-subarray-of-size-k5313/1			
Smallest Subarray with a given sum (easy)	https://leetcode.com/problems/minimum-size-subarray-sum/			
Longest Substring with K Distinct Characters (medium)	https://www.geeksforgeeks.org/problems/longest-k-unique-characters-substring0853/1			
Fruits into Baskets (medium)	https://leetcode.com/problems/fruit-into-baskets/			
No-repeat Substring (hard)	https://leetcode.com/problems/longest-substring-without-repeating-characters/			
Longest Substring with Same Letters after Replacement (hard)	https://leetcode.com/problems/longest-repeating-character-replacement/			
Longest Subarray with Ones after Replacement (hard)	https://leetcode.com/problems/max-consecutive-ones-ii/			
Minimum size subarray SUM	https://leetcode.com/problems/minimum-size-subarray-sum/			
Minimum size subarray (HARD)	https://leetcode.com/problems/minimum-window-substring/description/?envType=study-plan-v2&envId=top-interview-150			
Problem Challenge 1: Permutation in a String (hard)	https://leetcode.com/problems/permutation-in-string/			
Problem Challenge 2: String Anagrams (hard)	https://leetcode.com/problems/find-all-anagrams-in-a-string/			
Problem Challenge 3: Smallest Window containing Substring (hard)	https://leetcode.com/problems/minimum-window-substring/			
Problem Challenge 4: Words Concatenation (hard)	https://leetcode.com/problems/substring-with-concatenation-of-all-words/			
Pattern: Kadane pattern				
Maximum subarray sum	https://leetcode.com/problems/maximum-subarray/dim_source-chatgpt.com			
Minimum Subarray Sum	https://www.geeksforgeeks.org/problems/smallest-sum-contiguous-subarray/1			
Maximum product subarray	https://leetcode.com/problems/maximum-product-subarray/?utm_source=chatgpt.com			
Maximum subarray sum with one deletion	https://leetcode.com/problems/maximum-subarray-sum-with-one-deletion/description/			
Maximum absolute sum of any subarray	https://leetcode.com/problems/maximum-absolute-sum-of-any-subarray/			
Maximum sum in circular array variant	https://leetcode.com/problems/maximum-sum-circular-subarray/?utm_source=chatgpt.com			
4. Pattern: Merge Intervals				
Merge Intervals (medium)	https://www.educative.io/courses/grkking-the-coding-interview/3jVpK8ABxw			
Insert Interval (medium)	https://www.educative.io/courses/grkking-the-coding-interview/3jVNMjPEM			
Intervals Intersection (medium)	https://www.educative.io/courses/grkking-the-coding-interview/1EXVjRANQD			
Conflicting Appointments (medium)	https://www.geeksforgeeks.org/check-if-any-two-intervals-overlap-among-a-given-set-of-intervals/			
Problem Challenge 1: Minimum Meeting Rooms (hard)	https://www.linkedin.com/problems/meeting-rooms-ii/			
Problem Challenge 2: Maximum CPU Load (hard)	https://www.geeksforgeeks.org/maximum-cpu-load-from-the-given-list-of-jobs/			
Problem Challenge 3: Employee Free Time (hard)	https://www.codetrain.co/employee-free-time			
5. Pattern: Cyclic Sort				
Cyclic Sort (easy)	https://www.geeksforgeeks.org/sort-an-array-which-contains-1-to-n-values-in-on-using-cycle-sort/			
Find the Missing Number (easy)	https://leetcode.com/problems/missing-number/			
Find all Missing Numbers (easy)	https://leetcode.com/problems/find-all-numbers-disappeared-in-an-array/			
Find the Duplicate Number (easy)	https://leetcode.com/problems/find-the-duplicate-number/			
Find all Duplicate Numbers (easy)	https://leetcode.com/problems/find-all-duplicates-in-an-array/			
Problem Challenge 1: Find the Corrupt Pair (easy)	https://thecodingsimplified.com/find-corrupt-pair/			
Problem Challenge 2: Find the Smallest Missing Positive Number (medium)	https://leetcode.com/problems/first-missing-positive/			
Problem Challenge 3: Find the First K Missing Positive Numbers (hard)	https://thecodingsimplified.com/find-the-first-k-missing-positive-number/			
6. Pattern: In place Reversal of a LinkedList				
Reverse a LinkedList (easy)	https://leetcode.com/problems/reverse-linked-list/			
Reverse a Sub-list (medium)	https://leetcode.com/problems/reverse-linked-list-ii/			
Reverse every K-element Sub-list (medium)	https://leetcode.com/problems/reverse-nodes-in-k-group/			
Problem Challenge 1: Reverse alternating K-element Sub-list (medium)	https://www.geeksforgeeks.org/reverse-alternate-k-nodes-in-a-singly-linked-list/			
Problem Challenge 2: Rotate a LinkedList (medium)	https://leetcode.com/problems/rotate-list/			
7. Pattern: Stack				
Balanced Parentheses	https://leetcode.com/problems/valid-parentheses/description/			
Reverse a String				
Decimal to Binary Conversion				
Next Greater Element	https://leetcode.com/problems/next-greater-element-i/		https://leetcode.com/problems/next-greater-element-ii/	https://leetcode.com/problems/next-greater-element-iii/
Sorting a Stack				
Simplify Path	https://leetcode.com/problems/simplify-path/			
8. Pattern: Monotonic Stack				
Next Greater Element (easy)	https://leetcode.com/problems/next-greater-element-i/		https://leetcode.com/problems/next-greater-element-ii/	https://leetcode.com/problems/next-greater-element-iii/
Daily Temperatures (easy)	https://leetcode.com/problems/daily-temperatures/			
Remove Nodes From Linked List (easy)	https://leetcode.com/problems/remove-nodes-from-linked-list/			
Remove All Adjacent Duplicates in String (easy)	https://leetcode.com/problems/remove-all-adjacent-duplicates-in-string/			
Remove All Adjacent Duplicates in String II (medium)	https://leetcode.com/problems/remove-all-adjacent-duplicates-in-string-ii/			
Remove K Digits (hard)	https://leetcode.com/problems/remove-k-digits/			
9. Pattern: Hash Maps				
First Non-repeating Character (easy)	https://leetcode.com/problems/first-unique-character-in-a-string/			
Largest Unique Number (easy)	https://leetcode.com/problems/largest-unique-number/			
Maximum Number of Balloons (easy)	https://leetcode.com/problems/maximum-number-of-balloons/			
Longest Palindrome (easy)	https://leetcode.com/problems/longest-palindrome/			
Ransom Note (easy)	https://leetcode.com/problems/ransom-note/			
10. Pattern: Tree Breadth First Search				
Binary Tree Level Order Traversal (easy)	https://leetcode.com/problems/binary-tree-level-order-traversal/			
Reverse Level Order Traversal (easy)	https://leetcode.com/problems/binary-tree-level-order-traversal-ii/			
Zigzag Traversal (medium)	https://leetcode.com/problems/binary-tree-zigzag-level-order-traversal/			
Level Averages in a Binary Tree (easy)	https://leetcode.com/problems/average-of-levels-in-binary-tree/			
Minimum Depth of a Binary Tree (easy)	https://leetcode.com/problems/minimum-depth-of-binary-tree/			
Maximum Depth of a Binary Tree (easy)	https://leetcode.com/problems/maximum-depth-of-binary-tree/			
Level Order Successor (easy)	https://www.geeksforgeeks.org/level-order-successor-of-a-node-in-binary-tree/			
Connect Level Order Siblings (medium)	https://leetcode.com/problems/populating-next-right-pointers-in-each-node/			
Problem Challenge 1: Connect All Level Order Siblings (medium)	https://www.educative.io/learn/connect-all-siblings			
Problem Challenge 2: Right View of a Binary Tree (easy)	https://leetcode.com/problems/binary-tree-right-side-view/			
11. Pattern: Tree Depth First Search				
Binary Tree Path Sum (easy)	https://leetcode.com/problems/path-sum/			
All Paths for a Sum (medium)	https://leetcode.com/problems/path-sum-ii/			
Sum of Path Numbers (medium)	https://leetcode.com/problems/path-sum-iii/			
Path With Given Sequence (medium)	https://www.geeksforgeeks.org/check-root-to-leaf-path-given-sequence/			
Count Paths for a Sum (medium)	https://leetcode.com/problems/path-sum-iv/			
Problem Challenge 1: Tree Diameter (medium)	https://leetcode.com/problems/diameter-of-binary-tree/			
Problem Challenge 2: Path with Maximum Sum (hard)	https://leetcode.com/problems/binary-tree-maximum-path-sum/			
12. Pattern: Graphs				
Graph Traversal: Depth First Search (DFS)				
Graph Traversal: Breadth First Search (BFS)				
Find if Path Exists in Graph (easy)	https://leetcode.com/problems/find-if-path-exists-in-graph/			
Number of Provinces (medium)	https://leetcode.com/problems/number-of-provinces/			
Minimum Number of Vertices to Reach All Nodes (medium)	https://leetcode.com/problems/minimum-number-of-vertices-to-reach-all-nodes/			
13. Pattern: Island (Matrix traversal)				
Number of Islands (easy)	https://leetcode.com/problems/number-of-islands/			
Biggest Island (easy)				
Flood Fill (easy)	https://leetcode.com/problems/flood-fill/			
Number of Closed Islands (easy)	https://leetcode.com/problems/number-of-closed-islands/			
Problem Challenge 1 (easy)				
Problem Challenge 2 (medium)				
Problem Challenge 3 (medium)				
14. Pattern: Two Heaps				
Find the Median of a Number Stream (medium)	https://leetcode.com/problems/find-median-from-data-stream/			
Sliding Window Median (hard)	https://leetcode.com/problems/sliding-window-median/			
Maximize Capital (hard)	https://leetcode.com/problems/capital-gain/			
*Maximum Sum Combinations (medium)	https://www.interviewbit.com/problems/maximum-sum-combinations/			
15. Pattern: Subsets				
Subsets (easy)	https://www.educative.io/courses/grkking-the-coding-interview/g2OgbUwVG			
Subsets With Duplicates (easy)	https://www.educative.io/courses/grkking-the-coding-interview/7n3k3Y3QONv			
Permutations (medium)	https://www.educative.io/courses/grkking-the-coding-interview/88R83yn3KY			
String Permutations by changing case (medium)				
Balanced Parentheses (hard)				
Unique Generalized Abbreviations (hard)	https://leetcode.com/problems/generalized-abbreviation/			
16. Pattern: Modified Binary Search				
Order-agnostic Binary Search (easy)	https://www.geeksforgeeks.org/order-agnostic-binary-search/			
Ceiling of a Number (medium)	https://www.geeksforgeeks.org/ceiling-in-a-sorted-array/		https://www.geeksforgeeks.org/floor-in-a-sorted-array/	
Next Letter (medium)	https://leetcode.com/problems/find-smallest-letter-greater-than-target/			
Number Range (medium)	https://leetcode.com/problems/find-first-and-last-position-of-element-in-sorted-array/			

Search in a Sorted Infinite Array (medium)	https://www.geeksforgeeks.org/find-position-element-sorted-array-infinite-numbers/			
Minimum Difference Element (medium): Find the floor & ceil take the difference. minimum would be the ans	https://www.geeksforgeeks.org/find-the-maximum-element-in-an-array-which-is-first-increasing-and-then-decreasing/			
Bitonic Array Maximum (easy)	https://leetcode.com/problems/find-in-mountain-array/			
Problem Challenge 1: Search Bitonic Array (medium)	https://leetcode.com/problems/search-in-rotated-sorted-array/			
Problem Challenge 2: Search in Rotated Array (medium)	https://leetcode.com/problems/search-in-rotated-sorted-array/			
Problem Challenge 3: Rotation Count (medium)	https://www.geeksforgeeks.org/find-rotation-count-rotated-sorted-array/			
*Search a 2D Matrix (medium)	https://leetcode.com/problems/search-a-2d-matrix/			
*Minimum Number of Days to Make m Bouquets (medium)	https://leetcode.com/problems/minimum-number-of-days-to-make-m-bouquets/			
*Koko Eating Bananas (medium)	https://leetcode.com/problems/koko-eating-bananas/			
*Capacity To Ship Packages Within D Days (medium)	https://leetcode.com/problems/capacity-to-ship-packages-within-d-days/			
*Median of Two Sorted Arrays (hard)	https://leetcode.com/problems/median-of-two-sorted-arrays/			
17. Pattern: Bitwise XOR				
Single Number (easy)				
Two Single Numbers (medium)				
Complement of Base 10 Number (medium)				
Problem Challenge 1: Flip and Invert an Image (hard)				
18. Pattern: Top 'K' Elements				
Top 'K' Numbers (easy)	https://github.com/dipuj/Grokking-the-Coding-Interview-Patterns-for-Coding-Questions/blob/master/13-pattern-top-k-elements/02-top-k-numbers.md			
Kth Smallest Number (easy)				
K' Closest Points to the Origin (easy)	https://leetcode.com/problems/k-closest-points-to-origin/			
Connect Ropes (easy)				
Top 'K' Frequent Numbers (medium)				
Frequency Sort (medium)				
Kth Largest Number in a Stream (medium)	https://leetcode.com/problems/kth-largest-element-in-a-stream/			
K' Closest Numbers (medium)				
Maximum Distinct Elements (medium)				
Sum of Elements (medium)				
Rearrange String (hard)				
Problem Challenge 1: Rearrange String K Distance Apart (hard)				
Problem Challenge 2: Scheduling Tasks (hard)				
Problem Challenge 3: Frequency Stack (hard)				
19. Pattern: K-way merge	https://github.com/dipuj/Grokking-the-Coding-Interview-Patterns-for-Coding-Questions/blob/master/13-pattern-top-k-elements/13-HeapImplementation.md			
Merge K Sorted Lists (medium)	https://leetcode.com/problems/merge-k-sorted-lists/			
Kth Smallest Number in K Sorted Lists (Medium)	https://www.geeksforgeeks.org/find-m-th-smallest-value-in-k-sorted-arrays/			
Kth Smallest Number in a Sorted Matrix (Hard)	https://www.educative.io/courses/grokking-the-coding-interview/1N1VVKNgwz			
Smallest Number Range (Hard)	https://leetcode.com/problems/smallest-range-covering-elements-from-k-lists/			
Problem Challenge 1: K Pairs with Largest Sums (hard)				
20. Pattern: Greedy Algorithms				
Valid Palindrome II (easy)	https://leetcode.com/problems/valid-palindrome-ii/			
Maximum Length of Pair Chain (medium)	https://leetcode.com/problems/maximum-length-of-pair-chain/			
Minimum Add to Make Parentheses Valid (medium)	https://leetcode.com/problems/minimum-add-to-make-parentheses-valid/			
Remove Duplicate Letters (medium)	https://leetcode.com/problems/remove-duplicate-letters/			
Largest Palindromic Number (Medium)	https://leetcode.com/problems/largest-palindromic-number/			
Removing Minimum and Maximum From Array (medium)	https://leetcode.com/problems/removing-minimum-and-maximum-from-array/			
21. Pattern: 0/1 Knapsack (Dynamic Programming)				
0/1 Knapsack (medium)	https://www.geeksforgeeks.org/0-1-knapsack-problem-dp-10/			
Equal Subset Sum Partition (medium)	https://leetcode.com/problems/partition-equal-subset-sum/			
Subset Sum (medium)	https://www.geeksforgeeks.org/subset-sum-problem-dp-25/			
Minimum Subset Sum Difference (hard)	https://www.geeksforgeeks.org/partition-a-set-into-two-subsets-such-that-the-difference-of-subset-sums-is-minimum/			
Problem Challenge 1: Count of Subset Sum (hard)				
Problem Challenge 2: Target Sum (hard)				
22. Pattern: Backtracking				
Combination Sum (medium)	https://leetcode.com/problems/combination-sum/	https://leetcode.com/problems/combination-sum-ii/	https://leetcode.com/problems/combination-sum-iii/	
Word Search (medium)	https://leetcode.com/problems/word-search/	https://leetcode.com/problems/word-search-ii/		
Sudoku Solver (hard)	https://leetcode.com/problems/sudoku-solver/			
Factor Combinations (medium)	https://leetcode.com/problems/factor-combinations/			
Split a String Into the Max Number of Unique Substrings (medium)	https://leetcode.com/problems/split-a-string-into-the-max-number-of-unique-substrings/			
23. Pattern: Trie				
Implement Trie (Prefix Tree) (medium)	https://leetcode.com/problems/implement-trie-prefix-tree/			
Index Pairs of a String (easy)	https://leetcode.com/problems/index-pairs-of-a-string/			
Design Add and Search Words Data Structure (medium)	https://leetcode.com/problems/design-add-and-search-words-data-structure/			
Extra Characters in a String (medium)	https://leetcode.com/problems/extra-characters-in-a-string/			
Search Suggestions System (medium)	https://leetcode.com/problems/search-suggestions-system/			
24. Pattern: Topological Sort (Graph)				
Topological Sort (medium)	https://www.youtube.com/watch?v=cBFEN077b4			
Tasks Scheduling (medium)	https://leetcode.com/problems/course-schedule/			
Tasks Scheduling Order (medium)	https://leetcode.com/problems/course-schedule-ii/			
All Tasks Scheduling Orders (hard)	https://leetcode.com/problems/course-schedule-iii/			
Alien Dictionary (hard)	https://leetcode.com/problems/alien-dictionary/			
Problem Challenge 1: Reconstructing a Sequence (hard)	https://leetcode.com/problems/sequence-reconstruction/			
Problem Challenge 2: Minimum Height Trees (hard)	https://leetcode.com/problems/minimum-height-trees/			
25. Pattern: Union Find				
Redundant Connection (medium)	https://leetcode.com/problems/redundant-connection/	https://leetcode.com/problems/redundant-connection-ii/		
Number of Provinces (medium)	https://leetcode.com/problems/number-of-provinces/			
Is Graph Bipartite? (medium)	https://leetcode.com/problems/is-graph-bipartite/			
Path With Minimum Effort (medium)	https://leetcode.com/problems/path-with-minimum-effort/			
26. Ordered Set				
Merge Similar Items (easy)	https://leetcode.com/problems/merge-similar-items/			
132 Pattern (medium)	https://leetcode.com/problems/132-pattern/			
My Calendar I (medium)	https://leetcode.com/problems/my-calendar-i/	https://leetcode.com/problems/my-calendar-ii/		